Measure of Dispersion (aka. Variability)

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Probability to Statistics (Descriptive vs. Inferential)

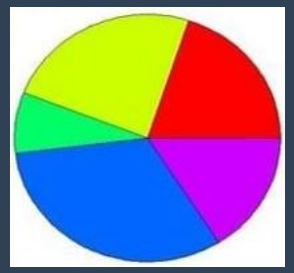
👉 Probability: A field of mathematics where we predicts the likelihood of future events. Probability is a number between 0 and 1, where a higher number means a more likely event.

👉 Statistics: A field of mathematics that analyzes the frequency of past events, which involves collecting, organizing, displaying, analyzing, interpreting, and presenting data.
There are two main types of statistics: descriptive statistics & inferential statistics

👉 Descriptive Statistics **: Describes various properties to measure & represent sample and population data.

👉 Statistical Inference: Statistical inference is based on probability and involves using a statistic to draw a conclusion about a parameter.

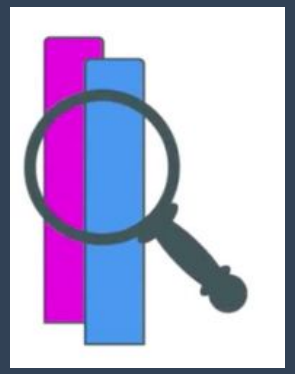
Statistics



- A statistic is a value that has been produced from a data collection, such as a summary measure, an estimate or projection.

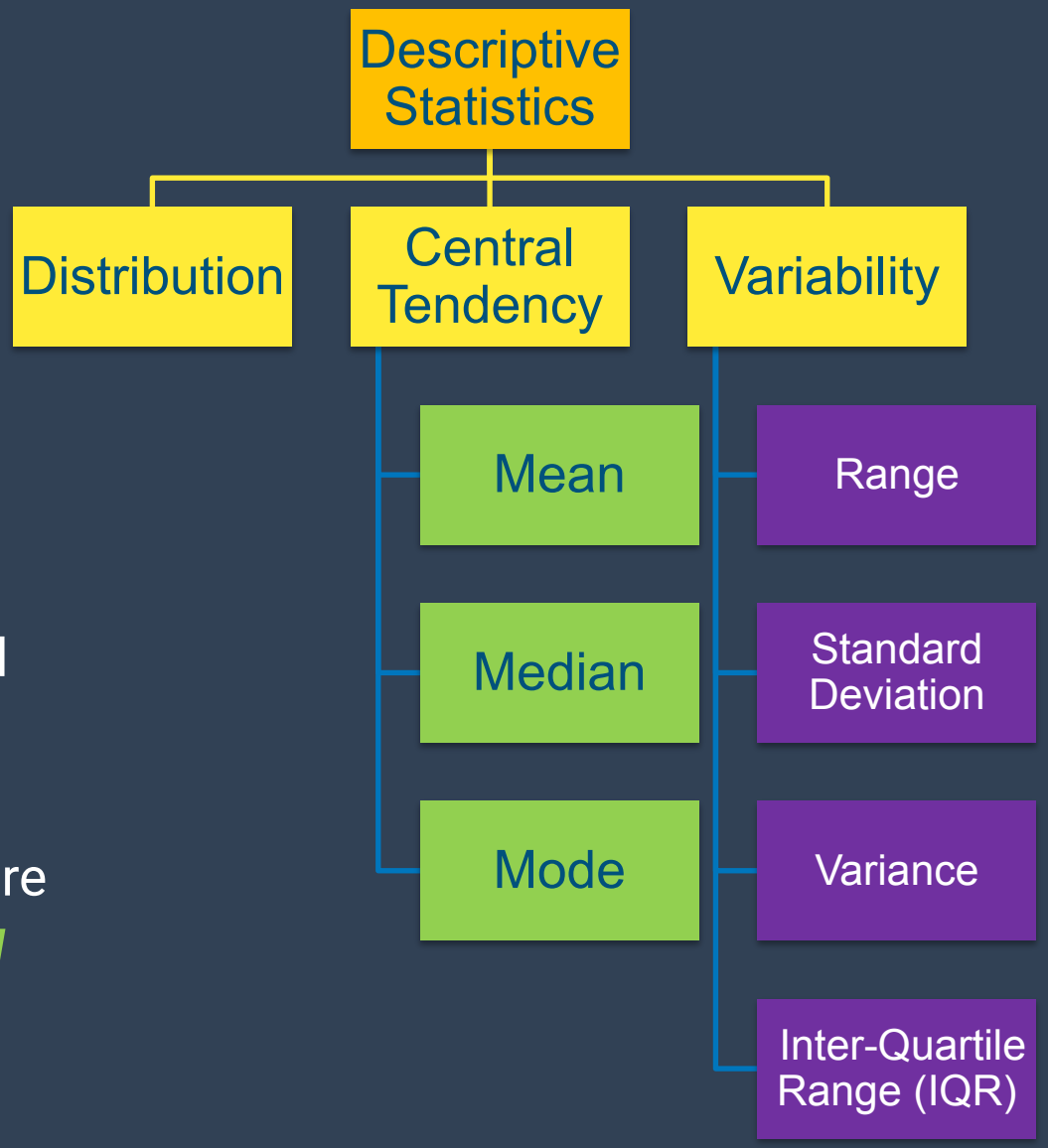
- Statistics is the science concerned with developing and studying methods for collecting, analyzing, interpreting and presenting empirical data.

- Statistical information is set of metrics that has been organized to serve a useful purpose.



Descriptive Statistics

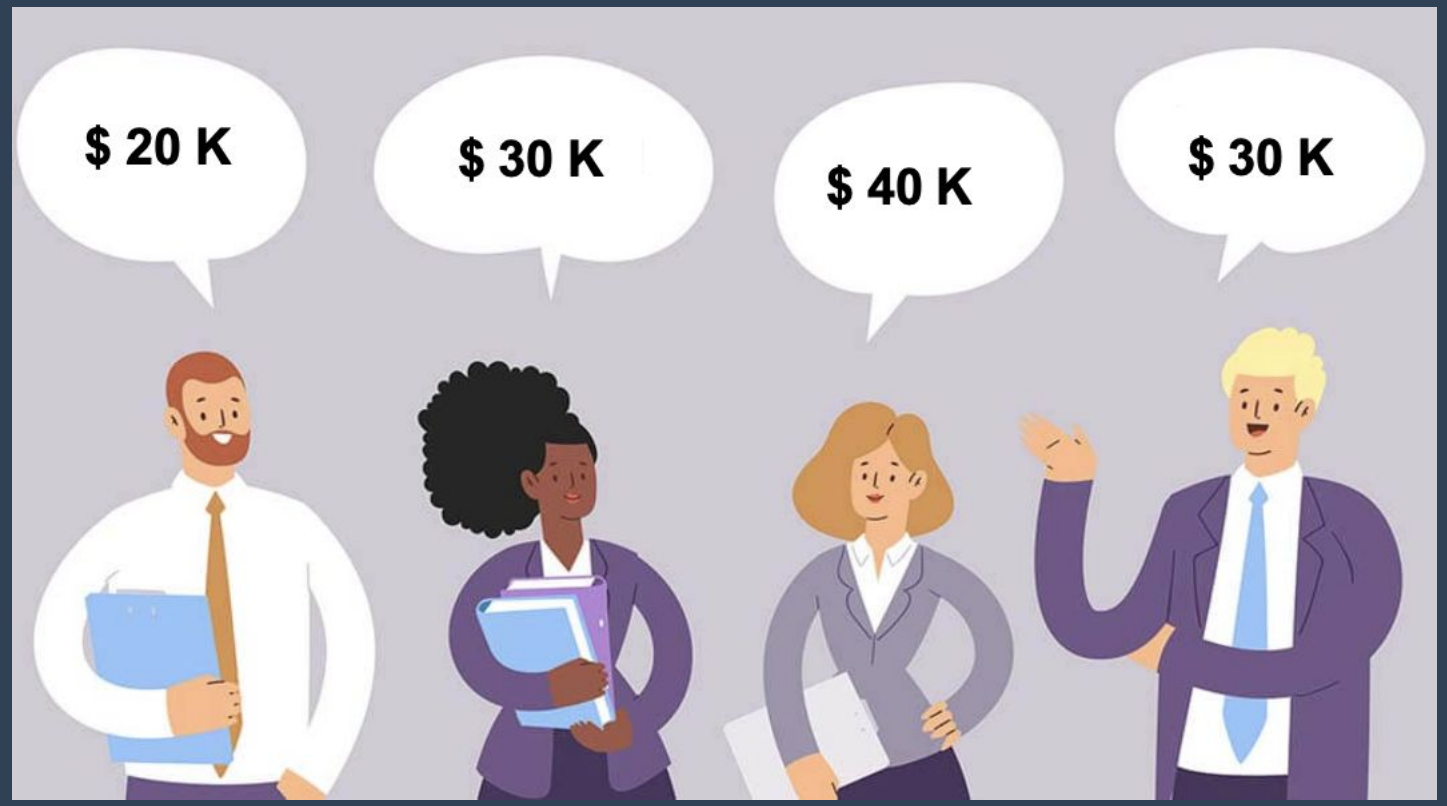
- Descriptive statistics summarizes or describes the characteristics of a data set.
- Descriptive statistics consists of three basic categories of measures: measures of central tendency, measures of variability (or spread) **, and frequency distribution.
- ***Measures of variability*** describe how “spread-out” are the values in the data set (***range, variance, standard deviation, IQR***).



Statistics

Statistics

- How to measure the **Dispersion *(variability)*** in the income of your friends?



Descriptive Statistics

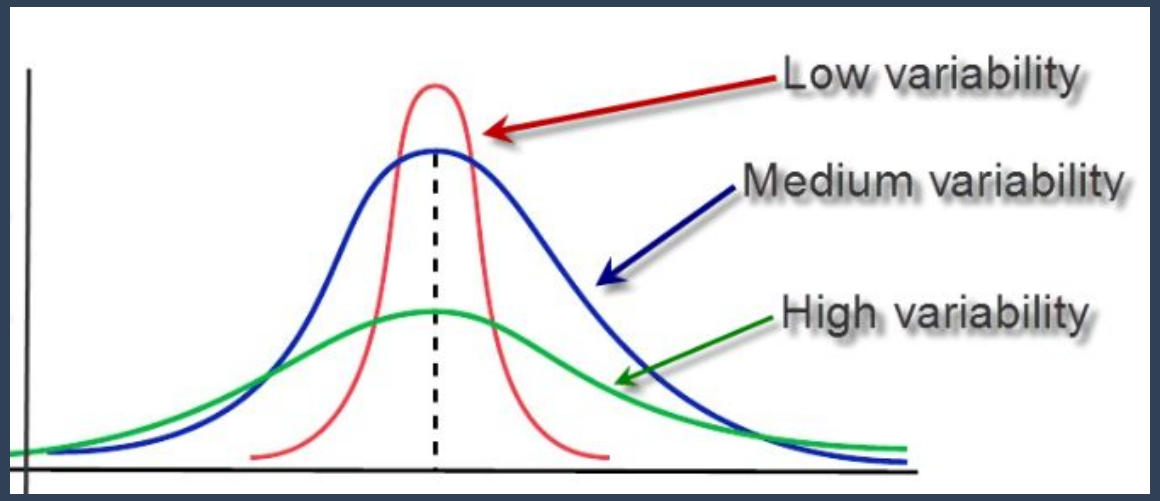
- How to summarize the **Variability** in the income of your friends in the dinner party ?

Variability

- While the central tendency, or average, tells us where most of the points lie, variability summarizes how far apart they are.

- Variability determines how well we can generalize results from the sample to the population.

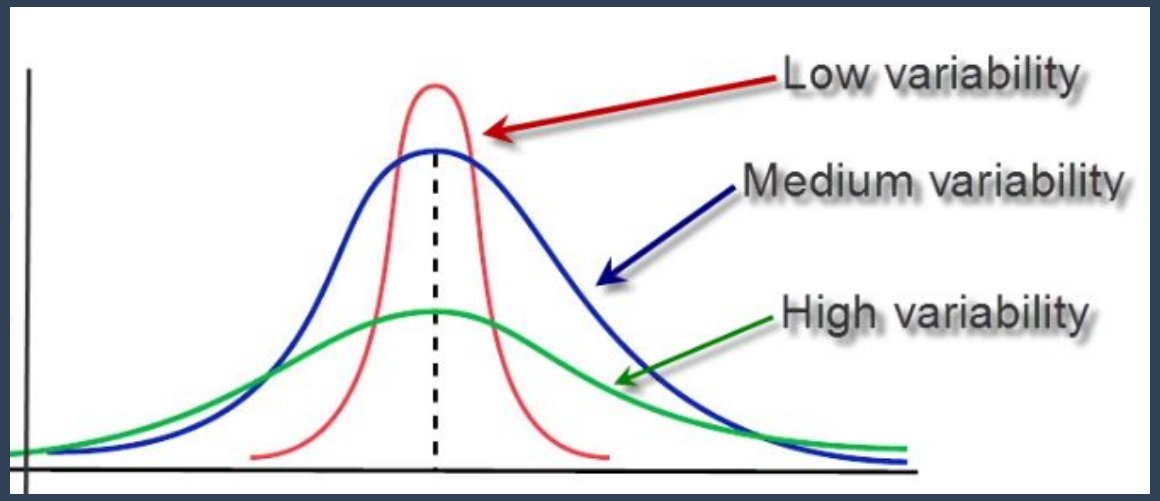
- Low variability means we can better predict information about the population based on sample data. High variability means that the values are less consistent, so it's harder to make predictions.



Variability

- Absolute Measures: The measures of dispersion which are expressed in terms of original units of a series are termed as Absolute Measures.

- Relative Measures: The measures of dispersion which are measured as a percentage or ratio of the average are termed Relative Measures. Relative Measures are sometimes known as coefficients of dispersion, without any specific unit attached to them.



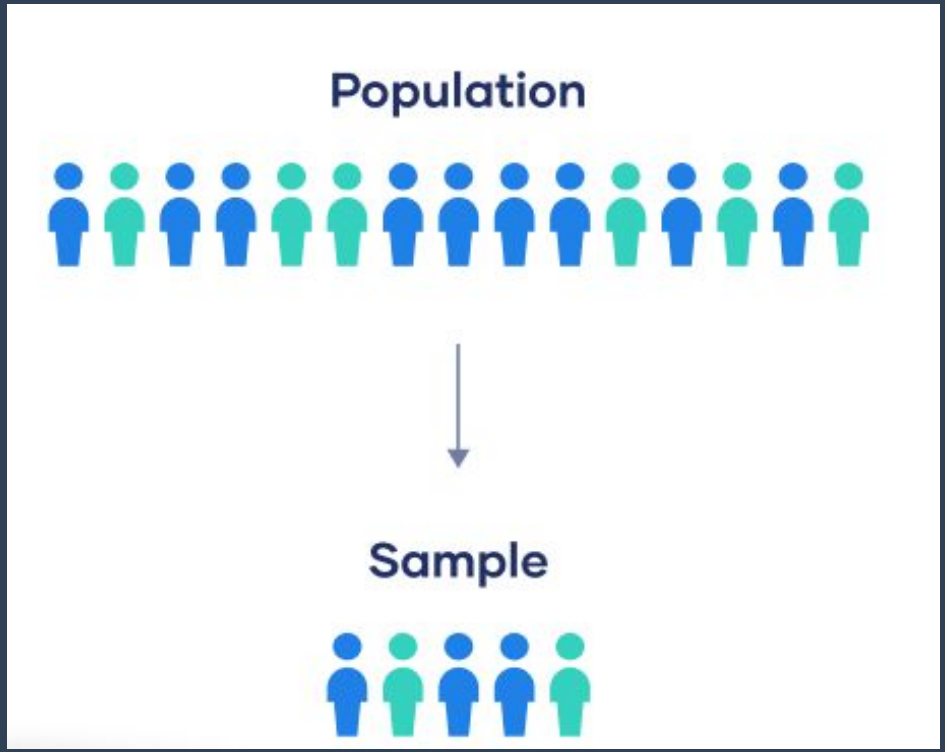
Various Measures of Dispersion

Range	The difference between the highest and lowest values
Variance	Average of squared distances from the mean
Standard-Deviation	Average distance from the mean
Interquartile Range (IQR)	The range of the middle half of a distribution

Population vs Sample

👉 A ***population*** is the entire group that you want to draw conclusions about.
E.g. Undergraduate students in the USA

👉 A ***sample*** is the specific group that you will collect data from. The size of the sample is always less than the total size of the population.
E.g. 300 undergraduate students from three US universities who volunteer for your AI research study



Range

Let's find out **Range**

Difference between the ***highest*** and ***lowest*** values signifies the ***spread*** of the data from the lowest to the highest value in the distribution.

Range => $40K - $20K

Range = $20K

Respondent	Base salary
Friend 1	$20 K
Friend 2	$30K
Friend 3	$40K
Friend 4	$30K

Variance

"How far are observations from the ***mean***"

$$\bar{X} = \text{mean of } X$$

Variance tells us the ***degree of spread*** in the data set. The more spread the data, the larger the variance is in relation to the mean.

Variance

The ***variance*** is a measure of ***variability***. It is calculated by taking the average of squared deviations from the mean.

$$\frac{\sum_{i=1}^{N}(X_i - \bar{X})^2}{N}$$

Respondent	Base salary
Friend 1	$20 K
Friend 2	$30K
Friend 3	$40K
Friend 4	$30K

Variance

$$\frac{\sum_{i=1}^{N}(X_i - \bar{X})^2}{N}$$

Mean = 30

Variance = **[(20-30)^2 + 2*(30-30)^2 + (40-30)^2] / 4**

= (100 + 2*0 + 100) / 4

= 200 / 4

= 50

Respondent	Base salary
Friend 1	$20 K
Friend 2	$30K
Friend 3	$40K
Friend 4	$30K

Unit of ***variance*** ?

Population vs Sample Variance

Population variance

$$\sigma^2 = \frac{\sum(X-\mu)^2}{N}$$

σ^2 = population variance

$\sum$ = sum of...

X = each value

μ = population mean

N = number of values in the population

Sample variance

$$s^2 = \frac{\sum(X-\bar{x})^2}{n-1}$$

s^2 = sample variance

$\sum$ = sum of...

X = each value

$\bar{x}$ = sample mean

n = number of values in the sample

Why ***(n-1)***?

Population vs Sample Variance

Population variance

$$\sigma^2 = \frac{\sum(X-\mu)^2}{N}$$

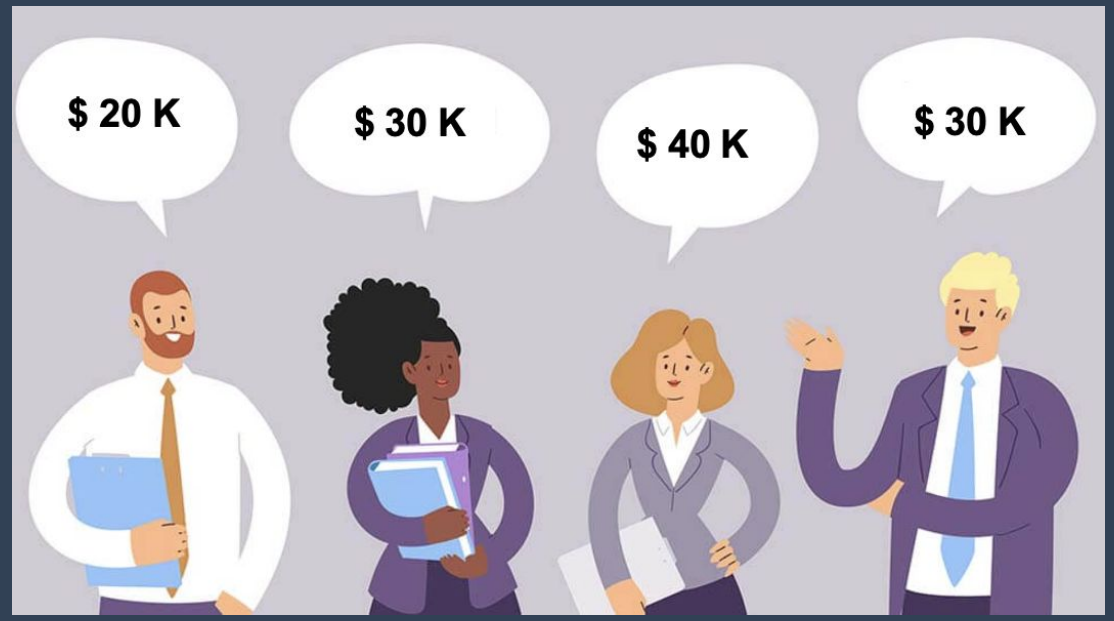
Sample variance

$$s^2 = \frac{\sum(X-\bar{x})^2}{n-1}$$

- Standard deviation calculated from the sample as an estimate of the standard deviation of the population.
- Because the observed values fall the standard deviation which is calculated using deviations from the sample mean underestimates the desired standard deviation of the population.
- Using ***N–1*** instead of ***N*** as the divisor corrects for that by making the result a little bit bigger.
- The correction has a larger proportional effect when ***N*** is small than when it is large.

Standard Deviation

The ***standard deviation*** is the average amount of variability in your dataset. It tells you, on average, how far each value lies from the mean.



$$\sqrt{\text{Variance}}$$

Standard Deviation

Make Variance Interpretable!

$$\sqrt{\text{Variance}}$$

Mean = 30

Variance = 50

Standard Deviation = ***Sqrt (Variance)*** = ***7.07***

Unit of ***Std. Dev.*** ?

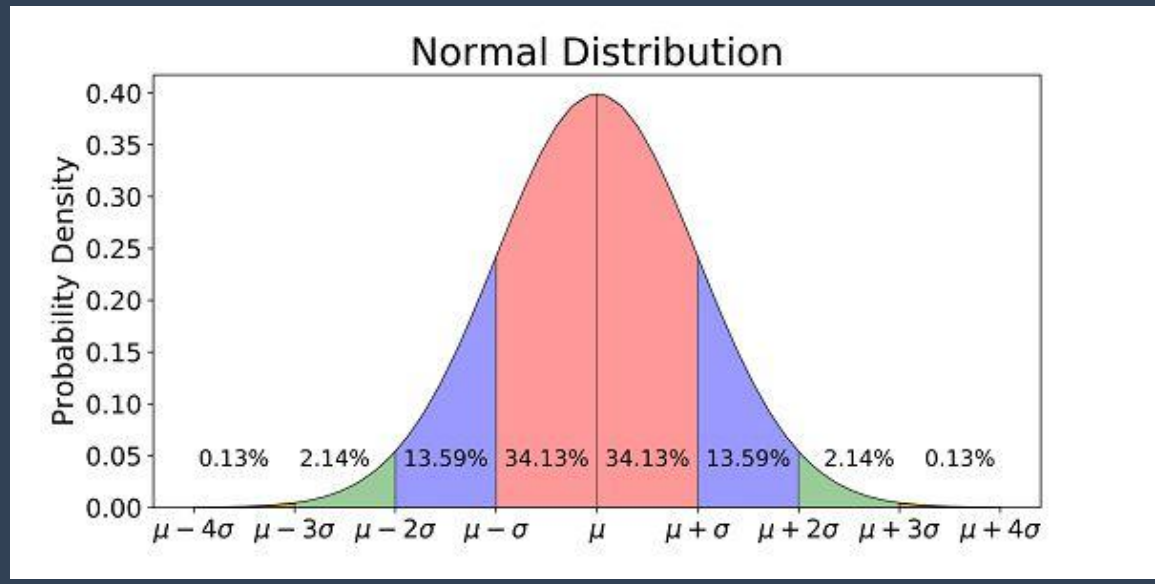
Respondent	Base salary
Friend 1	$20 K
Friend 2	$30K
Friend 3	$40K
Friend 4	$30K

Standard Deviation in Normal Distribution

The ***standard deviation*** and the ***mean*** together can tell us where most of the values in your frequency distribution lie if they follow a normal distribution.

The ***empirical rule of 68-95-99.7***:

- Around ***68%*** of scores are ***within 1 standard deviation*** of the mean,
- Around ***95%*** of scores are ***within 2 standard deviations*** of the mean,
- Around ***99.7%*** of scores are ***within 3 standard deviations*** of the mean.

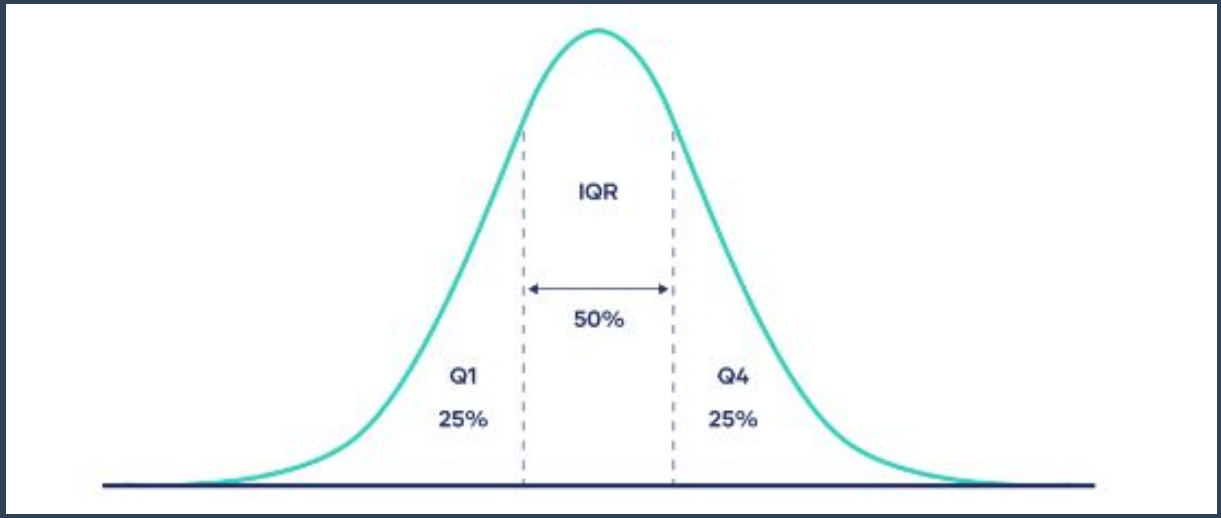


Six Sigma

Interquartile Range (IQR)

- The ***interquartile range*** gives us the spread of the middle in the data distribution.

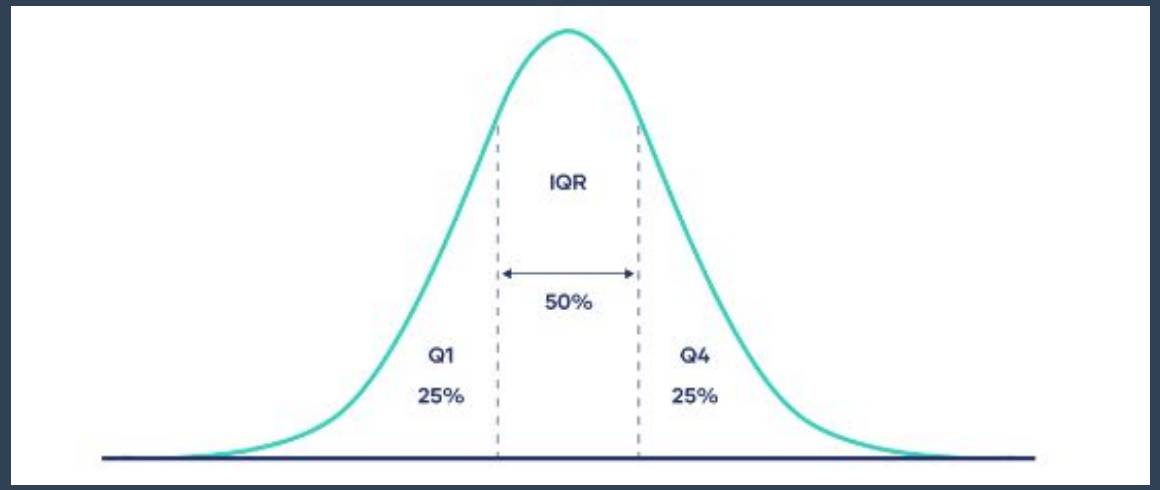
- For any distribution that's *ordered* from *low to high*, the interquartile range contains half of the values. While the *first quartile (Q1) contains the first 25% of values*, the *fourth quartile (Q4) contains the last 25% of values*.



Interquartile Range (IQR)

Let’s calculate the ***Interquartile Range***.

Interquartile Range (IQR)



Respondent	Base salary
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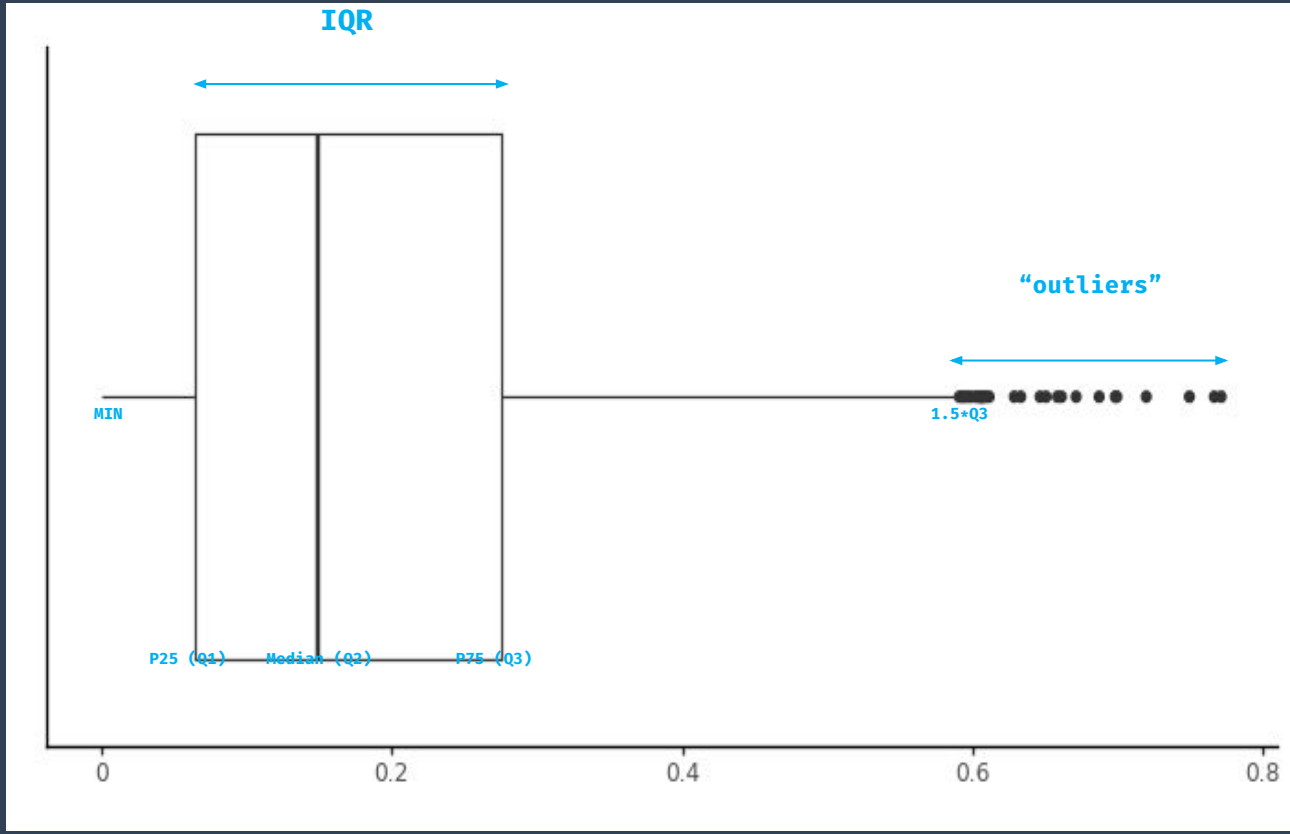
First Quartile (Q1) => 27.5

Third Quartile (Q3) => 32.5

IQR = ***Q1- Q3*** = ***5***

Boxplot

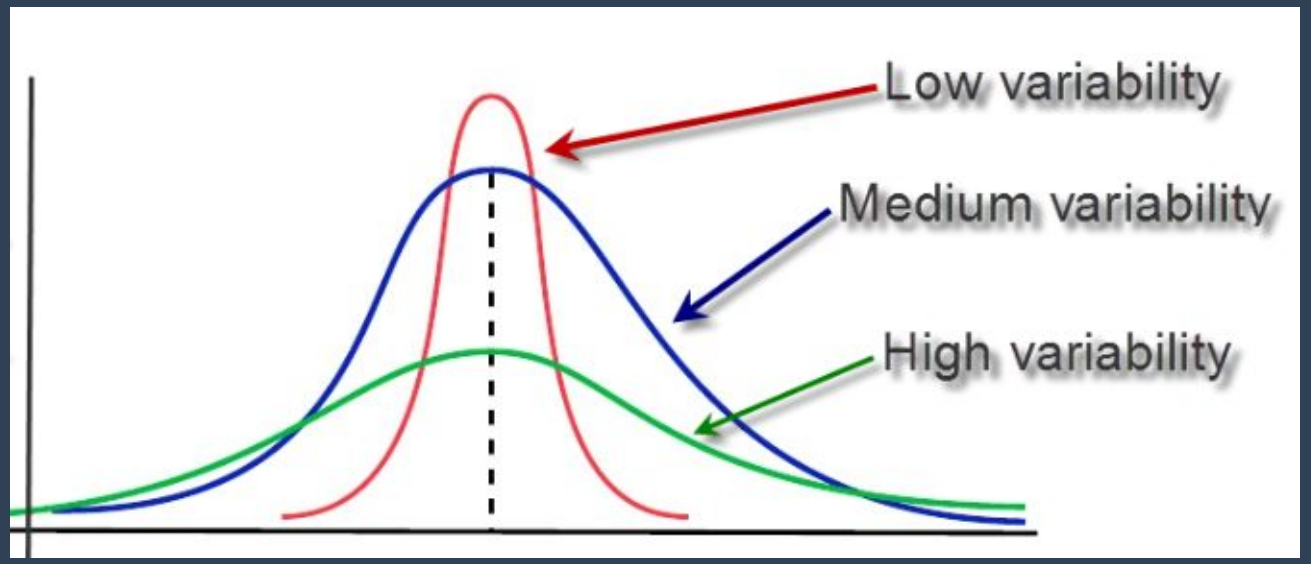
- A popular way to visualize IQR
- Many variations, but typically
  - Thick line representing median
  - Box over IQR
  - “Whiskers” extending to either [Q1 - 1.5×IQR, Q3 + 1.5×IQR] or most extreme observation
  - “***Outliers***” beyond whiskers as points
- Often used to formally define/identify “outliers” as: observations outside of [Q1 - 1.5×IQR, Q3 + 1.5×IQR]



Questions?

- When we need to adjust bias: Population OR Sample Variance/ Std Dev.?

- In presence of Outlier: Is median OR mean more impacted?



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- When we need to adjust bias: Population OR Sample Variance/ Std Dev.?
  - Sample Variance: Calculated with a divisor of N−1.

- In presence of Outlier: Is median OR mean more impacted?

Questions?

- When we need to adjust bias: Population OR Sample Variance/ Std Dev.?
  - Sample Variance: Calculated with a divisor of N−1.

- In presence of Outlier: Is median OR mean more impacted?
  - Mean: Outliers are extreme data values that differ from the rest of the data. The mean can be more impacted than the median by outliers, especially in skewed or outlier-containing distributions.

Time to code